

Queues and asynchronous work

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Caching and stale information

Without opening the old lesson, explain one key idea from Caching and stale information. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sandboxing and resource limits

Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Timeouts retries and cancellation

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Queues and asynchronous work. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Queues and asynchronous work. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Jobs A, B, C accepted; A completed, B failed, C pending. How many completed, and may success be claimed for all accepted?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Worker commits job J then crashes before queue acknowledgement. J is delivered again. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Queue ordering and exactly-once properties must be declared, not assumed from a simple mock. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Jobs A, B, C accepted; A completed, B failed, C pending. How many completed, and may success be claimed for all accepted?

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled queues and asynchronous work evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of queues and asynchronous work. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Queues and asynchronous work

Queued is not complete. A conversion service queues job J12 rather than processing it during submission.

States: queued, running, succeeded, failed. Submission returns J12 with queued status.

Queues and asynchronous work

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Accept request

J12 acknowledges receipt, not completed conversion.



2. Start worker

The worker marks J12 running before processing.



3. Publish checked result

Only a checked output permits succeeded status.

CHANGE ONE THING

The worker crashes before producing checked output.

Explain the changed result using the matching lesson.

Explain what the queued J12 submission response proves and what it does not. Which event permits succeeded status, and what status is honest after a worker crash?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Queued is not complete. A conversion service queues job J12 rather than processing it during submission.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only a checked output permits succeeded status.

1. Start worker
2. Publish checked result
3. Accept request

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Publish checked result Only a checked output permits succeeded status. Asynchronous work separates submission from completion. Honest status and a job identifier let users track that difference.

Practice 2: explain the deciding evidence

Changed-case practice: The worker crashes before producing checked output. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. The job is failed or awaiting a controlled retry, not succeeded.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The job is failed or awaiting a controlled retry, not succeeded. Receipt is not evidence that conversion finished.

Repair a missing building block

Review Caching and stale information (ai-6/21). Goal: Explain the mechanism and terminology of caching and stale information. Short practice: Identify the version mismatch in the cached 09:00 answer. What should the first version-5 request return, and what permits a later version-5 cache hit?. Return to this week and explain what you changed.

Review Sandboxing and resource limits (ai-6/19). Goal: Explain the mechanism and terminology of sandboxing and resource limits. Short practice: Check A against both resource limits and B against network isolation. Why must A stop after three seconds even if its output remains only 2 MB?. Return to this week and explain what you changed.

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the mechanism and terminology of timeouts retries and cancellation. Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
